

Vivek Chandrakant Bhave

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Education

M.S. in Artificial Intelligence & Data Science

2025 – 2027

Atal Bihari Vajpayee Indian Institute of Information Technology and Management (ABV-IIITM), Gwalior
CGPA (Semester 1): 9.32

Skills

Programming: C, C++, Python, SQL

Libraries: NumPy, Pandas, Matplotlib, Seaborn, Scikit-learn

Machine Learning: Regression, Classification, Clustering, Model Evaluation

Data Analysis: EDA, Data Cleaning, Feature Engineering, Data Visualization

Tools: Git, GitHub, Jupyter Notebook, VS Code

Projects

IMDB Sentiment Analysis Pipeline

Python, Scikit-learn, TF-IDF, Pandas, SQL, Git

- Built an end-to-end sentiment analysis pipeline on **40k IMDB reviews** including EDA, TF-IDF feature engineering, and classical ML models.
- Tuned TF-IDF parameters (**min_df**, **max_df**, **max_features**, **n-grams**) to reduce noise and improve classification accuracy.
- Trained and compared multiple models: **Logistic Regression (0.89)**, **SVM (0.89)**, **Naive Bayes (0.87)**, **Random Forest (0.85)**, **Decision Tree (0.73)**.
- Performed detailed metric analysis using **precision**, **recall**, **F1-score**, and **confusion matrix**.
- Conducted **failure analysis of misclassified reviews** on mixed sentiment, sarcasm, and long reviews.
- Analyzed **limitations of binary sentiment classification**.
- Compared **linear vs tree-based models** for high-dimensional TF-IDF features.
- Designed for real-world use cases like **large-scale review monitoring**.
- GitHub: imdb-sentiment-analysis-pipeline

Human-Relatedness Embedding

Python, GloVe, NumPy, Pandas, Scikit-learn

- Evaluated alignment of embeddings with human judgments using **SimLex-999** and **MEN**.
- Used cosine similarity on **GloVe (50D–300D)** and measured with **Pearson** and **R²**.
- Best result on MEN: **correlation = 0.74**, **R² = 0.55**.
- GitHub: Human-Relatedness-embedding

Achievements

- Solved 38 problems on LeetCode (Arrays, Strings, Sliding Window, Prefix Sum, Recursion, Binary Search) — Profile
- Public DSA repository in C++ — GitHub